Can a Model Read a Script It Has Never Seen?

A predictive tokenization law and zero-shot cross-script transfer for Brahmic scene-text recognition

Ritu Baskey · research summary · 3 July 2026 · every number here comes from a result file on disk and is reproducible; nothing is estimated or projected

ABSTRACT

Scene-text recognition — reading shop signs, banners and posters in photographs — is data-starved for Indian scripts: collecting labeled photographs for all nine Brahmic scripts is prohibitively expensive. This project builds a way around that bottleneck in three connected results. (1) A script-aware tokenization + confidence-fusion method lifts Bengali scene-text word accuracy from 52.4% to 64.6% and statistically ties a specialist model trained on roughly 1000× more data. (2) An empirical law: one pre-registered, computable property of a writing system (BPE fertility — how many subword tokens a standard tokenizer spends per written character unit) predicts, before any training, which tokenization wins on which script — correct on 8 of 9 Brahmic scripts, 90.9% under leave-one-script-out validation. (3) A shared "abugida pivot space" that maps all nine scripts into one output vocabulary, in which a model trained on eight scripts reads the ninth — a script it has never seen one real photograph of — at up to 19.8% exact-word accuracy (base model: 0.0%). A pre-registered 27-experiment run now in progress completes result (3) across all nine scripts.

1 Scripts and data attempted so far

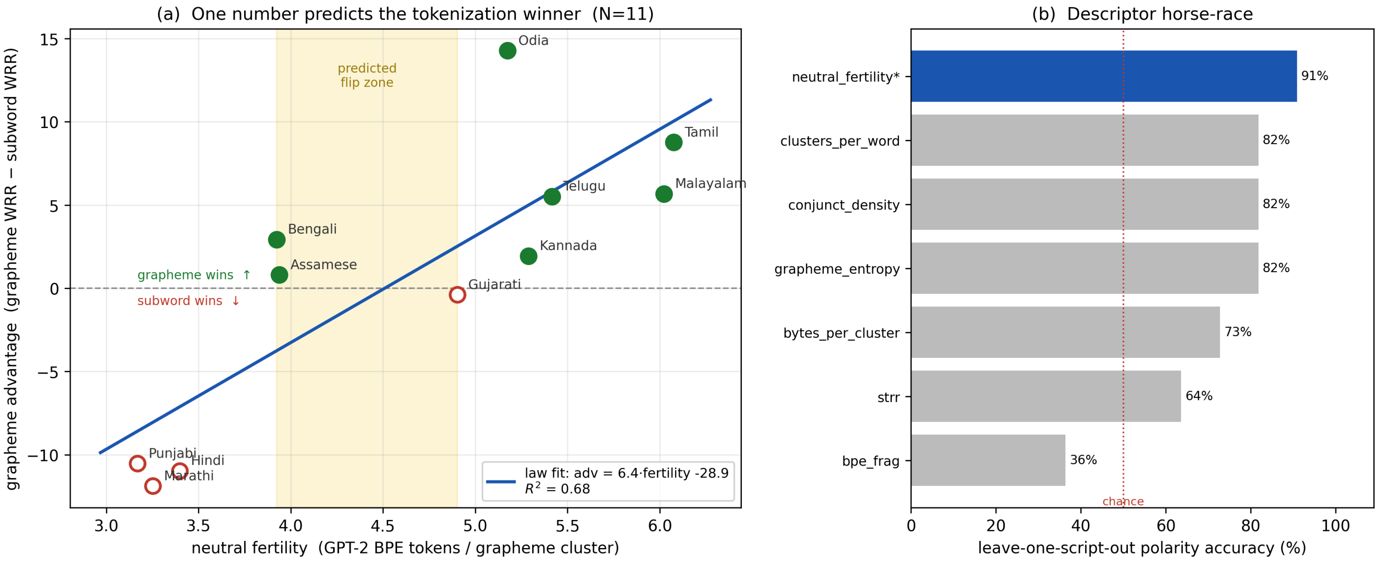
All nine Brahmic scripts have been trained and evaluated in the supervised study, via the 11 Indic languages of the public BSTD benchmark (IIT Jodhpur): Bengali script (Bengali + Assamese), Devanagari (Hindi + Marathi), Gurmukhi (Punjabi), Gujarati, Oriya (Odia), Tamil, Telugu, Kannada and Malayalam — plus English (Latin) as a control. Added to this: our own lab-annotated Bengali set (7,378 crops; I re-built its train/test split after finding photo-level leakage in the old one), and font-rendered synthetic data for every script. In total 44 fine-tuned model variants, all on a single shared RTX A5000. The zero-shot study (§4–5) covers the same nine scripts: Tamil and Telugu complete, Kannada in progress, six queued.

2 Result 1 — script-aware tokenization + fusion (the completed Bengali paper)

Base model Florence-2 (a small vision-language model), fine-tuned with LoRA. Injecting whole grapheme-cluster tokens (so a fused conjunct — two or three consonants written as one shape — is a single output unit) beats the stock BPE tokenizer; the two make different mistakes, and picking the more confident answer per image beats both. On our Bengali test set: Tesseract 24.8 → CRNN 44.1 → fine-tuned BPE 52.4 → best single model 62.2 → fusion 64.6% (+12.2 over standard fine-tuning). Replicated on public BSTD: Bengali 62.1, Assamese 38.5, Hindi 61.7 — on Hindi the polarity flips (BPE wins there), yet fusion still gains +4.8 ($p < 0.0001$). Head-to-head with BSTD's specialist PARSeq model (pre-trained on 8.48 million synthetic images): statistical tie on our independent test set (66.8 vs 64.6, $p = 0.50$) with ~1000× less training data. Model confidence genuinely predicts correctness (AUROC 0.88–0.91): in selective mode the system reaches 89.7% accuracy at 50% coverage. Full draft exists.

3 Result 2 — the tokenization-granularity law (why Hindi flips: it is predictable)

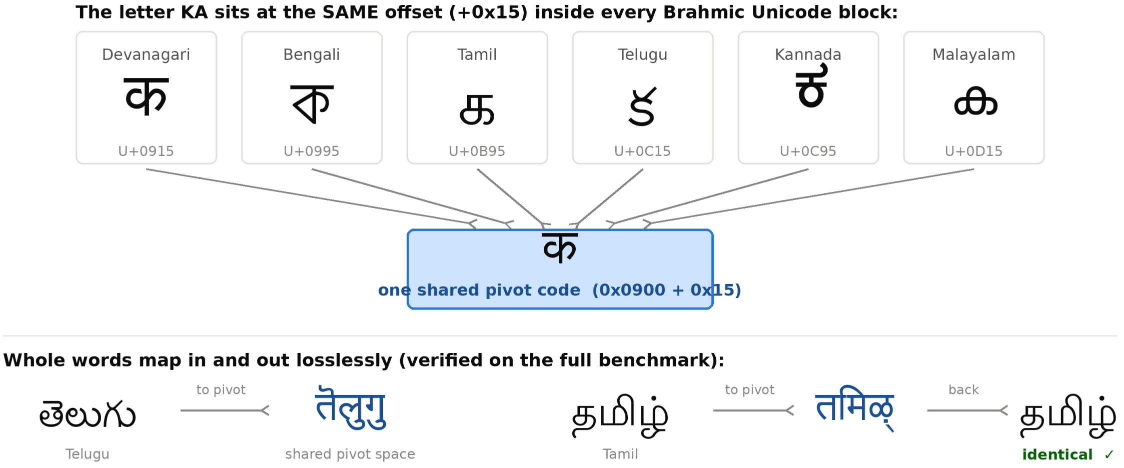
That polarity flip is not noise — it is the phenomenon. With hypotheses, descriptors and statistics frozen in a pre-registration on 18 June (before any result existed), all 11 languages were trained under identical budgets (1,620 images each) in both tokenizations:



One number predicts the winner: grapheme tokenization wins exactly where fertility is high (8/9 Brahmic scripts correct; the single miss, Gujarati, is a −0.39 near-tie). Under leave-one-script-out validation fertility scores 90.9% vs ≤ 82% for six competitor descriptors — including STRR, the predictor recent literature proposes instead, which degenerates to zero on every Indic script (itself a reportable finding). The fusion gain also grows with fertility (R^2 = 0.70; bootstrap CI on the slope excludes 0, pre-registered criteria met).

4 The shared representation: one "abugida pivot space" for all nine scripts

What is the shared representation? A deterministic, invertible Unicode offset map. Because the Brahmic scripts descend from a common ancestor, their Unicode blocks are aligned: the same sound sits at the same offset in every block. Shifting each character by a fixed per-script constant maps every script into one shared block (Devanagari's), losslessly:

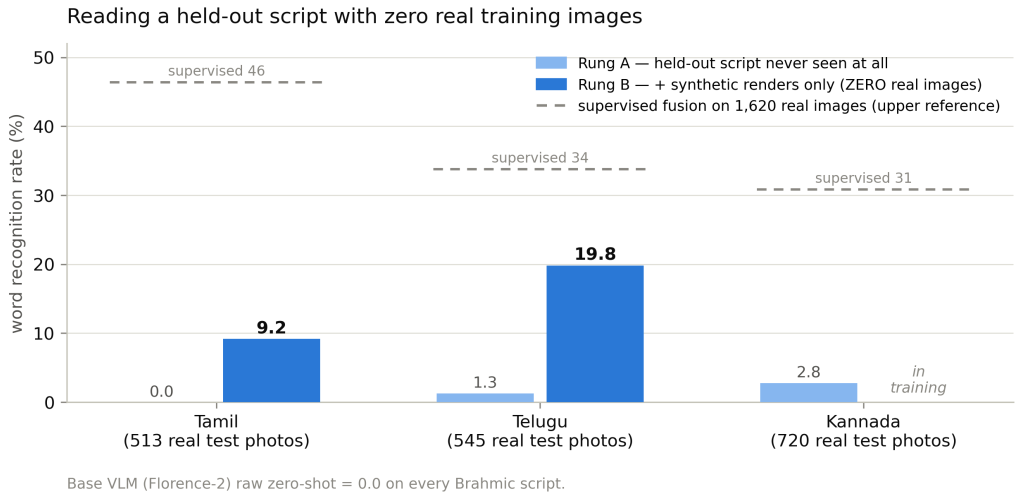


A model trained to output pivot text learns ONE output vocabulary shared by all scripts: a script it never saw still lands on output units it already knows from the other eight — the structural prerequisite for zero-shot transfer. Round-trip fidelity was verified word-by-word over the full benchmark before use.

5 Result 3 (in progress) — reading a held-out script with zero real images

Leave-one-script-out protocol, pre-registered: hold out one script entirely; train on 7,000 real photos of the ten other languages (in pivot space). Rung A adds nothing else — the held-out script is never seen in any form. Rung B additionally adds 3,240 synthetic font-rendered words of the held-out script — still zero real photographs of it. Evaluate exact-word accuracy on the held-out script's real test photos.

So, to answer directly: when Telugu is read at 19.8%, Telugu itself was the held-out script. That model was trained on Bengali, Assamese, Hindi, Marathi, Gujarati, Punjabi, Kannada, Malayalam, Odia and Tamil, plus computer-rendered Telugu only; it never saw a single real Telugu photograph, and was tested on 545 real ones. The base model reads them at 0.0%.



Transfer is real and large: +9.2 points on Tamil and +18.5 on Telugu over the no-exposure baseline, from synthetic renders alone — Telugu's zero-real-image model already reaches 59% of a supervised model trained on 1,620 real Telugu images (and 54% character accuracy). The honest remaining gap is synthetic-to-real domain shift, which the few-shot rung (adding just 50–100 real words) will measure next.

6 The falsifiable bet currently on the table

Which property predicts WHICH scripts transfer best — the law's fertility, or source-vocabulary coverage? On 2 July, with 7 of 9 outcomes still unknown even to us, ranked predictions under both were committed to the public git history (commit 047c30c); Malayalam is the decisive case (fertility's predicted best transferer, coverage's worst). Whichever loses is reported in full. A visual-similarity control descriptor was likewise committed result-blind (c5e7c28) to test the literature's strongest objection — "appearance, not structure, drives transfer" — with data.

7 Status and trajectory

5 of 27 pre-registered experiments are complete (a power outage on 30 June cost three days; the run resumes automatically when the shared GPU frees, fully crash-hardened). Next: the remaining 22 experiments (~1 GPU-week), the 9-script prediction-vs-transfer correlation, the few-shot rung, an extension toward 15–20 scripts. Plan: arXiv preprint mid-July; main paper to WACV 2027 (Round 2, 28 Aug); the completed Bengali paper to IJDAR / ICDAR 2027.

Integrity: hypotheses/descriptors/tests pre-registered before results (frozen 18 Jun 2026; amendments logged with git timestamps); train/test splits leakage-audited; pipeline audited end-to-end 26 Jun; negative results reported (Gujarati near-tie miss; ROVER fusion no better than max-confidence; STRR degenerate). All claims trace to result files in the repo.